

EXPLAINABLE AI-BASED SCREENING OF RETINOBLASTOMA

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Project Description

This project aims to detect Retinoblastoma from retinal fundus images using deep learning techniques. The system classifies images as Normal or Tumor and provides prediction results through a graphical user interface.

Technologies Used

- Python
- TensorFlow
- Keras
- OpenCV
- NumPy
- Tkinter

Dataset

- Retinal fundus images
- Classes: Normal and Tumor
- Preprocessing: Resizing, Noise Reduction, Normalization
- Augmentation: Rotation, Flip, Zoom, Shift, Shear, Brightness Adjustment

Algorithms Used

1. CNN
2. MobileNetV2

Project Workflow

1. Data Collection
2. Data Preprocessing
3. Data Augmentation
4. Model Training
5. Model Evaluation
6. GUI Development

7. Prediction Generation

How to Run the Project

1. Install required libraries.
2. Place the trained model (model.h5) in the project folder.
3. Run gui.py.
4. Upload a retinal image.
5. View the prediction result.

Output

The system predicts whether the uploaded retinal image is:

- Normal
- Tumor

Future Enhancements

- Larger dataset
- Multi-disease detection
- Web application deployment
- Improved accuracy using advanced models

Conclusion

The proposed system successfully detects Retinoblastoma using deep learning and provides an easy-to-use interface for prediction.

